

Roshanpreet Singh Dhiman

Full-Stack Developer · Frontend Engineer · Applied AI Developer

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[Portfolio](#) | [LeetCode](#)

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React.js, Node.js, Express.js, Electron.js, FastAPI, PyTorch

Tools & Platforms: Git, GitHub, VS Code, Xcode, MySQL, Firebase, Supabase, Ollama

EDUCATION

Chitkara University

B.E. Computer Science & Engineering

Rajpura, Punjab, India

Aug. 2025 – June 2029 (Expected)

FLAGSHIP PROJECTS

Autonomous AI Persona (Ada) | *Next.js, Supabase, Groq (Llama 3.3 70B), Vercel*

[GitHub](#) | [Live Demo](#)

- Engineered **Ada**, an autonomous AI security-researcher persona that ingests 500+ Hacker News posts per cycle via the Algolia API, scoring and summarizing threats through a Groq-hosted Llama 3.3 70B model on a cron-driven tick loop.
- Architected the full-stack Next.js + Supabase data pipeline in a 2-person team during a 48-hour hackathon, handling 1,000+ API calls per session; deployed and load-tested on Vercel with zero downtime.

Friday Desktop | *Electron.js, React, Vite, TypeScript, Python, FastAPI, Ollama*

[GitHub](#)

- Built a **fully offline, voice-controlled macOS assistant** with an Electron + React frontend and a FastAPI WebSocket backend, running Gemma 2 (2B) locally via Ollama — zero API keys required.
- Integrated faster-whisper speech-to-text and native macOS TTS across 7 runtime-parsed system commands (app launch, Spotify, screenshot, WhatsApp, file creation); rendered a **WebGL particle orb** with state-driven animations.

Mac Live Wallpaper | *Swift, AppKit, AVFoundation, Xcode*

[GitHub](#)

- Published a native macOS utility (Apple Silicon M1–M4, macOS 13+) using AVFoundation layers to loop any MP4 as a live desktop wallpaper at under 2% CPU overhead.
- Diagnosed and resolved an **EXC_BAD_ACCESS** crash caused by an AVFoundation player-layer teardown race condition and incorrect fullscreen-space detection.

SIDE PROJECTS

P95.AI | *React 18, Express.js, SQLite, Google Gemini API* — Top 5, ThinkRoot × Vortex 2026

[GitHub](#)

- Enriched **200+ B2B leads** from 6 platforms with a 5-factor ICP scoring model, auto-generating personalized cold emails and LinkedIn outreach for the top 50 prospects with A/B testing and 4-language i18n (EN/HI/ES/FR).

TriageAI | *React 19, TypeScript, Tailwind CSS, HTML5 Canvas, Web Audio API*

[GitHub](#)

- Developed a Level-1 Trauma ER triage dashboard with 60 fps animated ECG/SpO₂ waveforms and a rule-based patient-severity classifier, compiled into a zero-dependency 289 KB offline HTML bundle.

Fearless Invest (Feargon) | *Vanilla JS, Node.js, Supabase, Gemini 2.5 Flash*

[GitHub](#)

- Constructed a fintech SaaS simulator with a Gemini 2.5 Flash advisor that interprets market movements and tailors advice to individual risk profiles via CoinGecko & Yahoo Finance APIs, with Supabase-backed auth and portfolio persistence.

NagarSeva | *React, Node.js* — Hackathon Project

[GitHub](#) | [Live Demo](#)

- Delivered a civic-issue reporting platform enabling citizens to log, upvote, and track municipal complaints across 10+ categories with real-time status updates.

Invisibility Cloak | *Python, OpenCV, MediaPipe*

[GitHub](#)

- Programmed a real-time computer-vision app that captures the background frame and uses fist-gesture detection to replace the user silhouette with the stored background at 30 fps in a live camera feed.

GesturePlay AI | *Vanilla JS, HTML5 Canvas, Google MediaPipe, Web Audio API*

[GitHub](#) | [Live Demo](#)

- Created 6 controller-free arcade games using Google MediaPipe hand-landmark detection with progressive difficulty scaling and a zero-dependency Web Audio oscillator engine.

Network Stress-Test Simulator | *Python*

[GitHub](#)

- Implemented a network stress-testing script in Python to study denial-of-service attack patterns and defense strategies within a controlled lab environment.

PureSight | *Next.js, Tailwind CSS, Framer Motion, GSAP, IoT*

[GitHub](#)

- Prototyped an IoT water-safety platform that renders real-time water-quality sensor data through a Next.js dashboard with GSAP/Framer Motion scroll-triggered data visualizations.

AirSteer | *Python, OpenCV, MediaPipe*

[GitHub](#)

- Automated racing-game controls by mapping real-time webcam hand-tracking to steering, throttle, and brake keystrokes, with an on-screen HUD overlay.

GitStatCard | *Python, SVG, GitHub API*

[GitHub](#)

- Shipped a tool that generates data-driven, embeddable GitHub-stats SVG cards for any username to power custom developer README profiles.

Background Remover | *Python, OpenCV*

[GitHub](#)

- Launched a Python CV tool that detects and removes image backgrounds, exporting transparent PNGs from any input photo.

Open-Source Contributor — merged patches into **Zulip** (10K+ stars) and **Matplotlib** (18K+ stars), fixing bugs across two large-scale Python codebases.

ACHIEVEMENTS

Hackathon Results: Finalist at Jackathon; Top 5 at ThinkRoot × Vortex 2026 and Salesforce Crosswalk; selected participant at GRID 8.0.

LeetCode: 150+ problems solved, 100-Days Badge, 176 active days with a 74-day max streak.